

$$一. \int_0^1 \int_{\frac{x}{2}}^{\frac{x}{2} + \ln(1+\frac{y}{x})} dx dy$$

$$\text{令 } u = x+y, v = x, |J|:$$

$$I = \int_0^1 \int_{\frac{u}{2}}^u \frac{u \ln \frac{u}{v}}{\sqrt{1-u}} dv du$$

$$x=v, y=u-v$$

$$\therefore J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1$$

$$0 \leq u \leq 1, 0 \leq v \leq u.$$

$$I = \int_0^1 \int_{\frac{u}{2}}^u \frac{u \ln \frac{u}{v}}{\sqrt{1-u}} dv du$$

$$= \int_0^1 du \int_0^1 \frac{u (u-v \ln v)}{\sqrt{1-u}} dv$$

$$= \int_0^1 du \left[\frac{u^2 \ln u}{\sqrt{1-u}} + \frac{u^2}{\sqrt{1-u}} - \frac{u^2 \ln u}{\sqrt{1-u}} \right]$$

$$= \int_0^1 \frac{u^2}{\sqrt{1-u}} du \quad 1-u=t^2 \quad u=1-t^2$$

$$\text{令 } \sqrt{1-u}=t, \text{ 则 } u=0 \text{ 时, } t=1, u=1 \text{ 时, } t=0$$

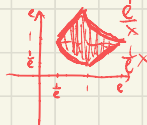
$$I = \int_1^0 \frac{(1-t^2)^2}{t} d(1-t^2)$$

$$= \int_1^0 -2(1-t^2)^2 dt$$

$$= 2 \int_0^1 (1+t^4 - 2t^2) dt$$

$$= 2 \left(t + \frac{1}{5} t^5 - \frac{2}{3} t^3 \right) \Big|_0^1 = 2 \left[1 + \frac{1}{5} - \frac{2}{3} \right] = \frac{16}{15}$$

$$二. |\ln x| + |\ln y| = 1$$



将积分区域分成两部分

$$S = \int_{\frac{1}{e}}^1 \left(ex - \frac{1}{ex} \right) dx + \int_1^e \left(\frac{e}{x} - \frac{x}{e} \right) dx$$

$$= \left(\frac{e}{2} x^2 - \frac{1}{e} \ln x \right) \Big|_{\frac{1}{e}}^1 + \left(e \ln x - \frac{1}{2e} x^2 \right) \Big|_1^e$$

$$= \frac{e}{2} - \left(\frac{e}{2} \times \frac{1}{e^2} + \frac{1}{e} \right) + e - \frac{1}{2e} e^2 + \frac{1}{2e}$$

$$= \frac{e}{2} - \frac{1}{2e} - \frac{1}{e} + e - \frac{e}{2} + \frac{1}{2e}$$

$$= e - \frac{1}{e}$$

三. 选择题

1. D 2. C 3. C 4. B

5. D 6. D 7. B 8. A

11.

$$(1) \begin{cases} z = 3x^2 + y^2 \\ z = 1 - x^2 \end{cases}$$

$$3x^2 + y^2 \leq z \leq 1 - x^2$$

$$4x^2 + y^2 = 1$$

$$y = \sqrt{1 - 4x^2}$$

$$x \in [-\frac{1}{2}, \frac{1}{2}]$$

$$\therefore \text{I} = \int_{-\frac{1}{2}}^{\frac{1}{2}} dx \int_{-\sqrt{1-4x^2}}^{\sqrt{1-4x^2}} dy \int_{3x^2+y^2}^{1-x^2} f(x,y,z) dz$$

$$(2) \begin{cases} z = x^2 + y^2 \\ y = x^2 \end{cases}, y=1, z=0$$

$$0 \leq z \leq x^2 + y^2 \quad x^2 \leq y \leq 1$$

$$-1 \leq x \leq 1$$

$$\therefore \text{I} = \int_{-1}^1 dx \int_{x^2}^1 dy \int_0^{x^2+y^2} f(x,y,z) dz$$

Ex (1) $\iiint_{\Omega} \frac{dx dy dz}{(x+y+z)^3}$ $\Omega: x=0, y=0, z=0, x+y+z=1$

$$0 \leq x \leq 1, \quad 0 \leq z \leq 1-y-x, \quad 0 \leq y \leq 1-x$$

$$\begin{aligned} \text{I} &= \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} \frac{dz}{(x+y+z)^3} \\ &= -\frac{1}{2} \int_0^1 dx \int_0^{1-x} \frac{1}{(x+y+z)^2} dy \\ &= -\frac{1}{2} \int_0^1 dx \int_0^{1-x} \left[\frac{1}{4} - \frac{1}{(x+y)^2} \right] dy \\ &= -\frac{1}{2} \int_0^1 \left[\frac{yx}{4} + \frac{1}{x+y} \right]_{y=0}^{y=1-x} dx \\ &= \frac{\ln 2}{2} - \frac{5}{16} \end{aligned}$$

$$(2) \iiint_{\Omega} y \cos(x+z) dx dy dz$$

$$y = \sqrt{x}, y=0, z=0, x+z = \frac{z}{2}$$

$$0 \leq z \leq \frac{z}{2} - x, \quad 0 \leq x \leq \frac{z}{2}, \quad \text{as } y \leq \sqrt{x}$$

$$\begin{aligned} \text{I} &= \int_0^{\frac{z}{2}} dx \int_0^{\sqrt{x}} dy \int_0^{\frac{z}{2}-x} y \cos(x+z) dz \\ &= \int_0^{\frac{z}{2}} dx \int_0^{\sqrt{x}} dy \left[y \sin(x+z) \right]_{z=0}^{\frac{z}{2}-x} \\ &= \int_0^{\frac{z}{2}} dx \int_0^{\sqrt{x}} (y - y \sin x) dy \\ &= \int_0^{\frac{z}{2}} dx \left(\frac{1}{2} y^2 - \frac{1}{2} y^2 \sin x \right)_{y=0}^{\sqrt{x}} \\ &= \int_0^{\frac{z}{2}} \left(\frac{1}{2} x - \frac{1}{2} x \sin x \right) dx \\ &= \frac{1}{4} x^2 \Big|_0^{\frac{z}{2}} - \frac{1}{2} \int_0^{\frac{z}{2}} x \sin x dx \\ &= \frac{1}{4} x^2 \Big|_0^{\frac{z}{2}} + \frac{1}{2} \int_0^{\frac{z}{2}} x \cos x dx \\ &= \frac{x^2}{16} + \frac{1}{2} x \cos x \Big|_0^{\frac{z}{2}} - \frac{1}{2} \int_0^{\frac{z}{2}} \cos x dx \\ &= \frac{x^2}{16} + 0 - 0 - \frac{1}{2} \sin x \Big|_0^{\frac{z}{2}} \\ &= \frac{z^2}{16} - \frac{1}{2} (1 - 0) = \frac{z^2 - 8}{16} \end{aligned}$$

Ex (1) $\iiint_{\Omega} xy dv$ $x^2 + y^2 = 1, z=0, z=1, x=0, y=0$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$r \in (0, 1), \quad \theta \in (0, 2\pi)$$

$$\begin{aligned} \text{I} &= \int_0^2 \cos \theta \sin \theta d\theta \int_0^1 r^3 dr \int_0^1 dz \\ &= \int_0^2 \cos \theta \sin \theta d\theta \left(\frac{1}{4} \right) \\ &= \frac{1}{8} \int_0^2 \sin \theta d\theta \\ &= \frac{1}{16} - \cos \theta \Big|_0^2 = -\frac{1}{16} \times (-1 - 1) = \frac{1}{8} \end{aligned}$$

$$(2) \iiint (x^2 + y^2) dv$$

$$\text{d. } x^2 + y^2 = 2z, z = 2$$

$$x^2 + y^2 = 4$$

$$x = r \cos \theta, y = r \sin \theta$$

$$0 \leq r \leq 2, 0 \leq \theta \leq 2\pi$$

$$z = \frac{x^2 + y^2}{2} = \frac{r^2}{2}$$

$$z \in \left[\frac{r^2}{2}, 2 \right]$$

$$I_{xy} = \int_0^{2\pi} d\theta \int_0^2 r^3 dr \int_{\frac{r^2}{2}}^2 dz$$

$$= \int_0^{2\pi} d\theta \int_0^2 r^3 dr \left(2 - \frac{r^2}{2} \right)$$

$$= \int_0^{2\pi} d\theta \int_0^2 r^3 \left(2 - \frac{r^2}{2} \right) dr$$

$$= \int_0^{2\pi} d\theta \int_0^2 r^2 \left(2 - \frac{r^2}{2} \right) d\left(\frac{r^2}{2}\right)$$

$$= 2 \int_0^{2\pi} d\theta \int_0^2 x(2-x) dx$$

$$= 2 \int_0^{2\pi} d\theta \left(x^2 - \frac{1}{2} x^3 \right) \Big|_0^2$$

$$= 2 \int_0^{2\pi} \left(4 - \frac{8}{3} \right) d\theta = 2 \times \frac{8}{3} \pi = \frac{16}{3} \pi$$